

ETHICS & RESPONSIBLE AI

Building Fair, Transparent, Safe, and Trustworthy Artificial Intelligence

AI BIAS & FAIRNESS

AI systems may unintentionally favor or disadvantage certain groups due to biases in training data.

Sources of Bias:

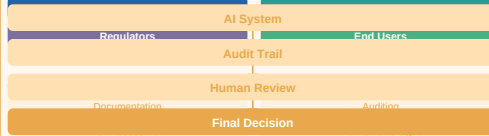


Real-World Examples:

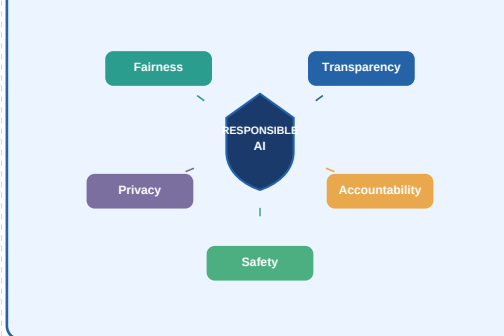
- Hiring algorithm bias
- Loan approval disparities
- Facial recognition gaps

ACCOUNTABILITY & GOVERNANCE

Who is Responsible?



RESPONSIBLE AI FRAMEWORK

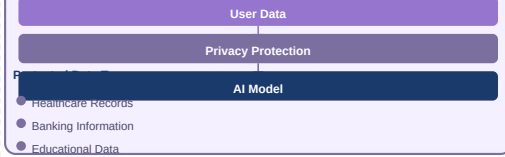


PRIVACY & DATA PROTECTION

Responsible AI must safeguard personal data and respect individual privacy rights throughout the AI lifecycle.

Privacy Techniques:

- Data Anonymization
- Differential Privacy
- Federated Learning
- Secure Storage



EXPLAINABILITY & TRANSPARENCY

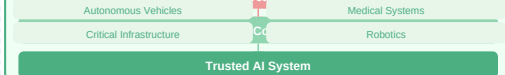


SAFETY & TRUSTWORTHY AI

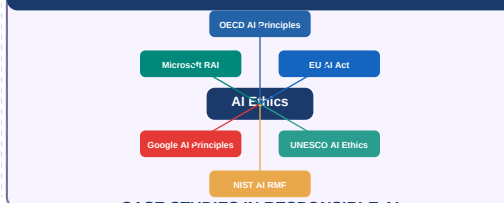
Key Characteristics:

- Reliability
- Robustness
- Security
- Human Control
- Ethical Alignment

Applications:



GLOBAL AI ETHICS FRAMEWORKS



CASE STUDIES IN RESPONSIBLE AI

Facial Recognition Bias	AI Recruitment Bias	Social Media Privacy	Healthcare AI XAI
Problem: Misidentified minorities 35% more often	Problem: Algorithm discriminated against female candidates	Problem: User data harvested without consent	Problem: Black-box diagnosis rejected by clinicians
Solution: Diverse training datasets + fairness testing	Solution: Audit & remove gender-correlated features	Solution: Federated learning + explicit consent models	Solution: SHAP-based explainability integration
Lesson: Representation in data is critical	Lesson: Audit pipelines before deployment	Lesson: Privacy by design is essential	Lesson: Trust requires transparency in medicine

KEY EVOLUTION OF AI ETHICS

